

Visual Doorbell Product Manual

Version 1



Tech for Good

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GOOD



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INTRODUCTION

Product Description

WHO?

Isolated seniors living in public rental flats, especially those who:

- Have partial or full hearing loss
 - Move slowly due to age or mobility impairments
 - Live alone and rely on community support
-

WHAT?

A **visual alert doorbell system** that provides:

1. **Clear visual indication inside the home when the doorbell is pressed**, and
 2. **An exterior e-paper display** that informs visitors through visual messages such as:
 - *"I'm coming" when the home owner presses a button to acknowledge*
-

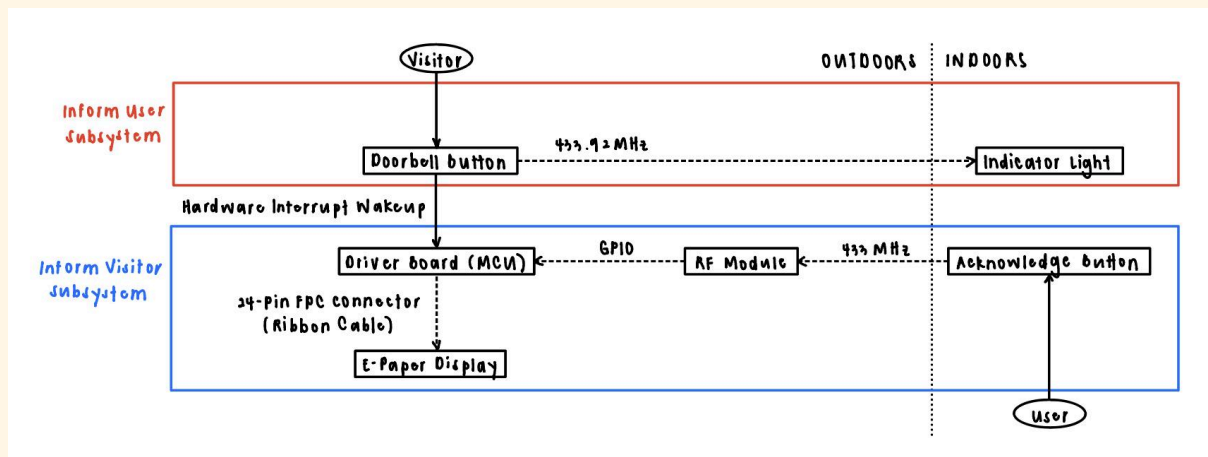
HOW?

This product augments an off-the-shelf wireless doorbell with an acknowledgement button and a responsive display to communicate with visitors.

Product Overview



System Diagram



The system is divided into 2 functional subsystems: an Inform User subsystem and an Inform Visitor subsystem. When a visitor presses the outdoor doorbell button, an indoor indicator light is activated to notify the user, and a hardware interrupt wakes the driver board (MCU). If the user is at home and chooses to press the acknowledge button, a 433 MHz wireless signal is transmitted and received by the RF module that is connected to the driver board via GPIO. The MCU will then update the E-Paper display through a ribbon cable to inform the visitor that the user is coming to the door with the message “I’m coming”.

Features

- Low-Power Operation**
 The MCU remains in deep-sleep mode during idle periods and is only woken by a hardware interrupt when the doorbell button is pressed, significantly reducing power consumption.
- No Wi-Fi or Internet Dependency**
 The system operates independently of Wi-Fi or cloud services, using local wired connections and 433 MHz RF communication for reliable operation.
- Accessible Visual Notifications**
 Visual indicators and an e-paper display provide clear feedback to both users and

visitors, making the system suitable for individuals with hearing loss or in noise-sensitive environments.

- **Reliable Short-Range Wireless Communication**

433 MHz RF communication ensures robust indoor–outdoor signalling with low power consumption and minimal configuration.

- **Simple and Intuitive Interaction**

The system uses only two buttons—one for the visitor and one for the user—minimising user effort and learning curve.

Bill of Materials

Item	Supplier/Product Link	Price	Quantity
ESP32-C6 module	ESP32-C6 Module (Seeed Studio)	\$5.20 USD	1
ePaper Driver board	ePaper Driver (Seeed Studio)	\$4.90 USD	1
ePaper Display	ePaper Display (Seeed Studio)	\$6.50 USD	1
LiPo Battery	LiPo Battery (Shopee)	\$6.43 SGD	1
3D Printed Casing	N/A	N/A	1
Wireless Doorbell	Wireless doorbell (Shopee)	\$10.19 SGD	1
Wireless receiver Button	Wireless Receiver Button (Shopee)	\$3.58 SGD	1
RF Receiver and Transmitter Module DIY Kit 433Mhz Super Outlier RF Module	433 Mhz RF module (Shopee)	\$3.05 SGD	1
Push Button	Push Button (Shopee)	\$1.98 SGD	1
Jumper Wires	Wires (Shopee)	\$1.64 SGD	5

USER GUIDE

Instructions for Use

Set Up

Setting up the outdoor Visual Doorbell	
1. Ensure that the driver board in the 3D housing is switched on.	
2. Ensure power source is properly connected.	
3. Tape off old doorbell.	

4. Mount a new doorbell button next to the existing doorbell. (Ensure the doorbell is placed clearly visible to visitors and protected from direct rain and excessive moisture.)

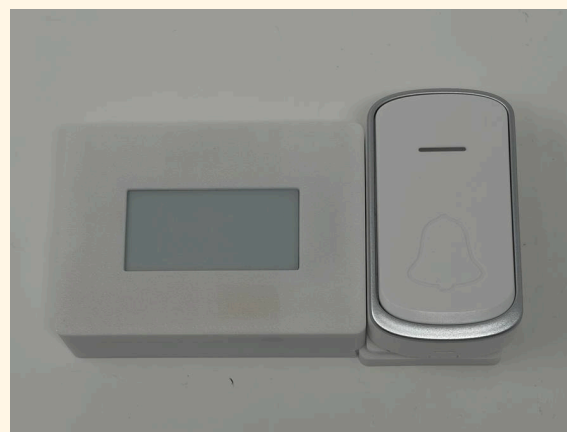


Setting up indoor acknowledgement system

1. Mount the indoor display unit to a power socket in a commonly used area of the home.
2. Ensure power source is properly connected and turned on.



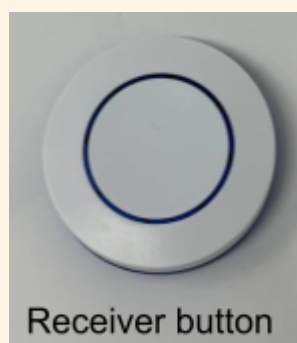
3. Press the doorbell button once to test the system



4. Confirm that the indoor display turns on when the doorbell is pressed



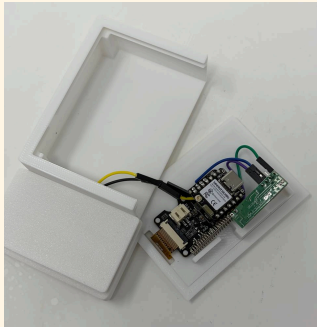
5. Confirm that the LCD display shows a message when the receiver button is pressed.



Maintenance

1. Recharge the battery when required by using a USB-C charger.
2. Keep the display clean and dry.
3. Check wiring connections periodically.

Troubleshooting

Problem	Possible Cause	What You Can Try
Display doesn't turn on	Battery is low	Recharge or replace the doorbell's battery fully.
	Driver board in the 3D housing is not switched on	 Check if the switch is shifted towards 'ON'.
	Loose connection of the battery/power source	Check the connection of the rechargeable battery. 
	Loose connection of the hardware interrupt button	Dismantle the casing and check the connection to the display (black and yellow wires). 

No audio response when doorbell button is pressed	Doorbell button is out of battery	Detach the doorbell from the system and remove the back lid. Change the coin battery inside the doorbell.
	Indoor display unit not set up properly	Ensure that the power socket is on.
Display message unclear	Screen not refreshed	Open the box and check all wires are securely connected.

MAKER GUIDE

Assembly Guide

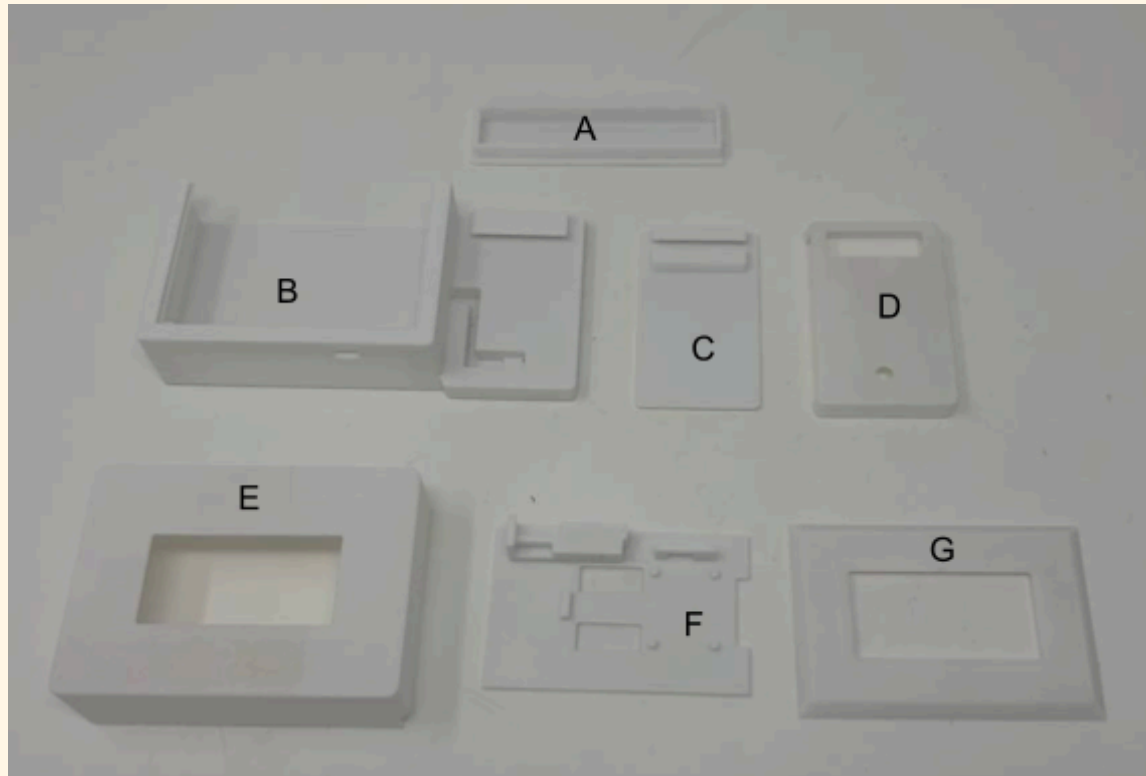
Off-the-Shelf Doorbell Adaptation

Parts Overview

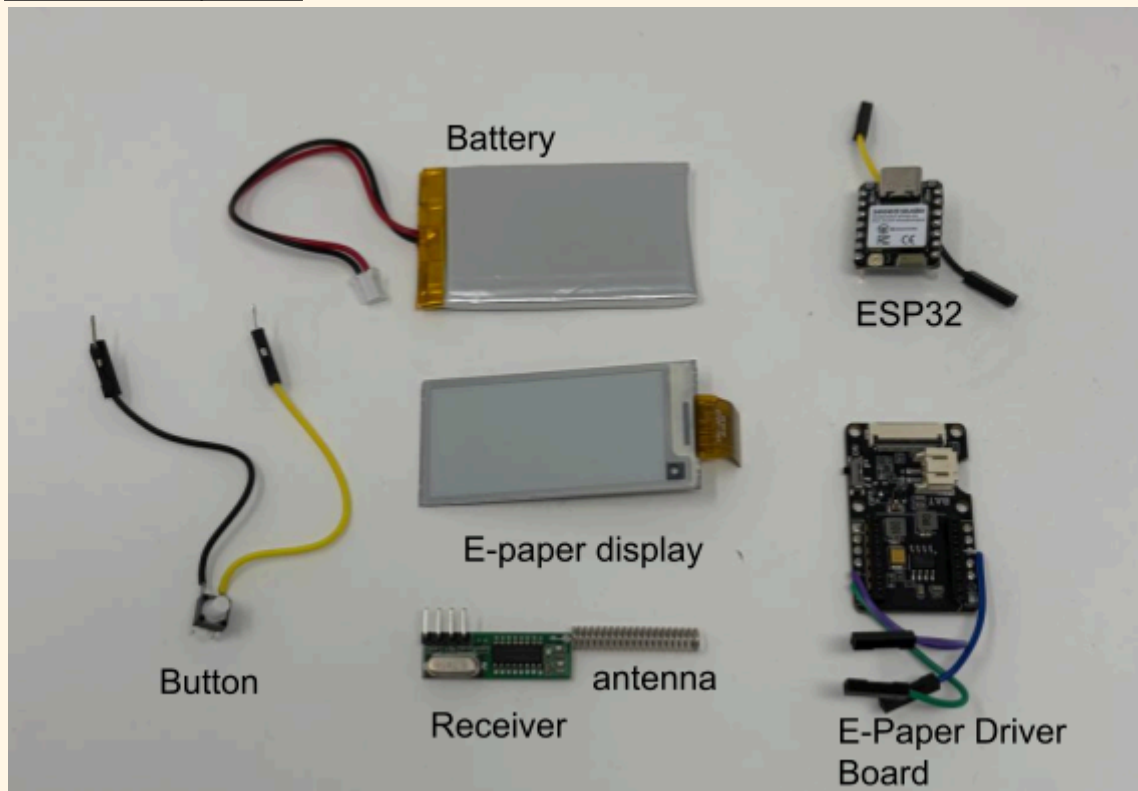
External parts



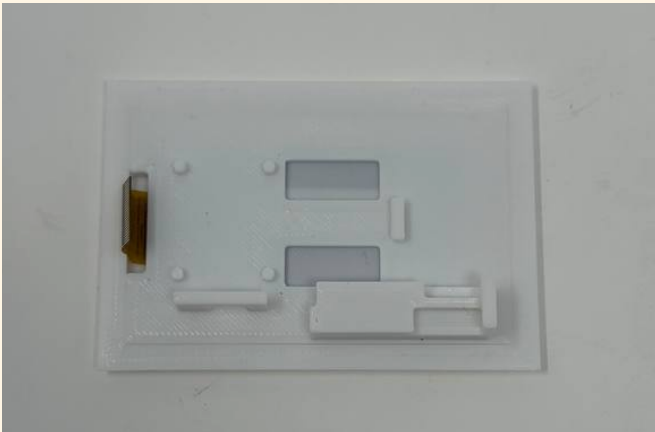
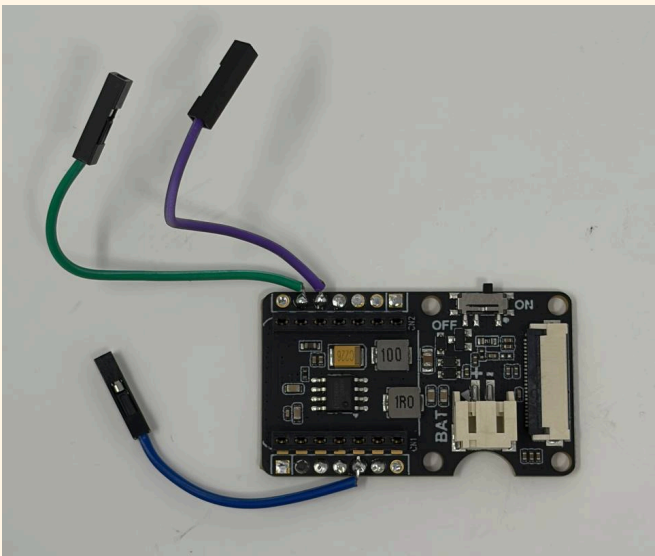
3D Printed Parts



Electronic Components



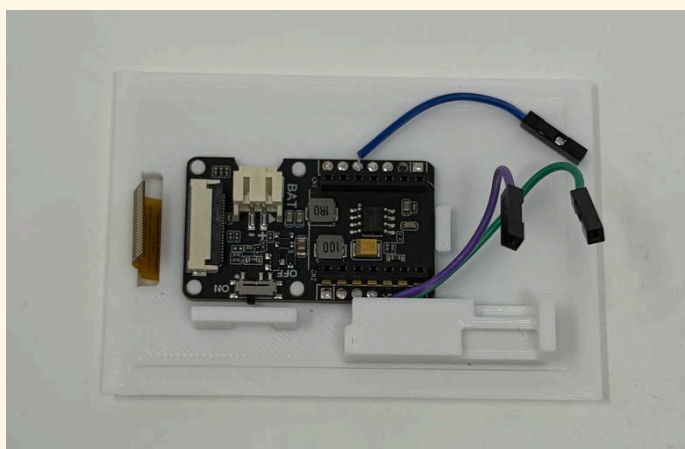
Set Up

Step	Diagram
1. Place the E-paper Display into the 3D printed Part G .	
2. Press and clip in the 3D printed cover Part F , to secure the E-paper display.	
3. Solder three wires to pins D4 (blue), GND (green) and 3V3 (purple) on the ePaper Driver Board.	

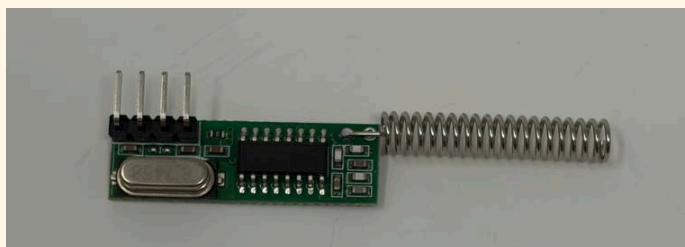
4. Solder two wires into the ESP module's pins GND (black) and MTMS (yellow).



5. Position and clip the ePaper Driver Board into the 3D printed **Part H**.

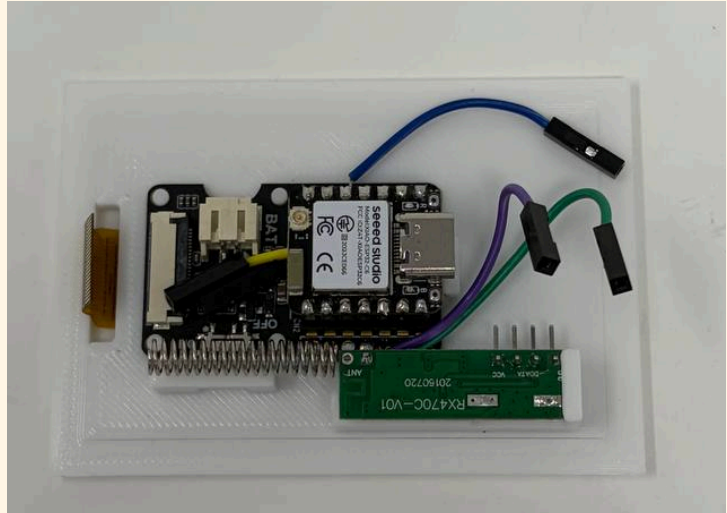


6. Solder antenna coil to 433 MHz RF receiver.



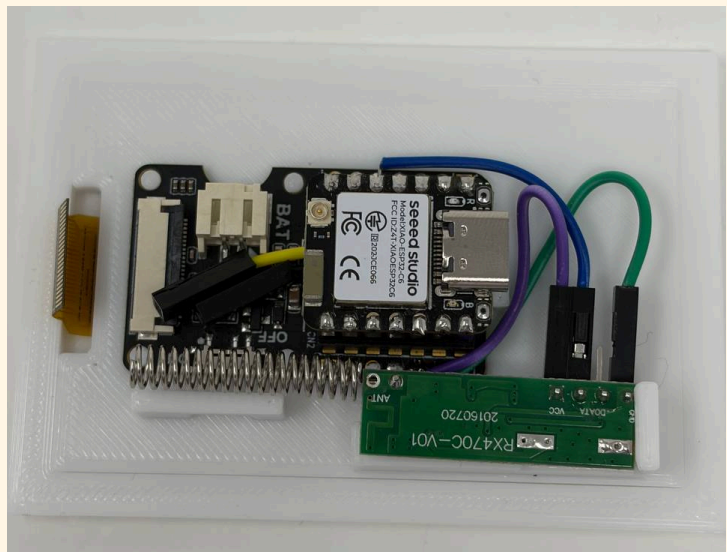
7. Clip the 433 MHz RF receiver into the bottom right corner of Part H (as seen in the picture).

Attach the ESP module with the USB-C port facing towards the receiver pins of the ePaper driver board.

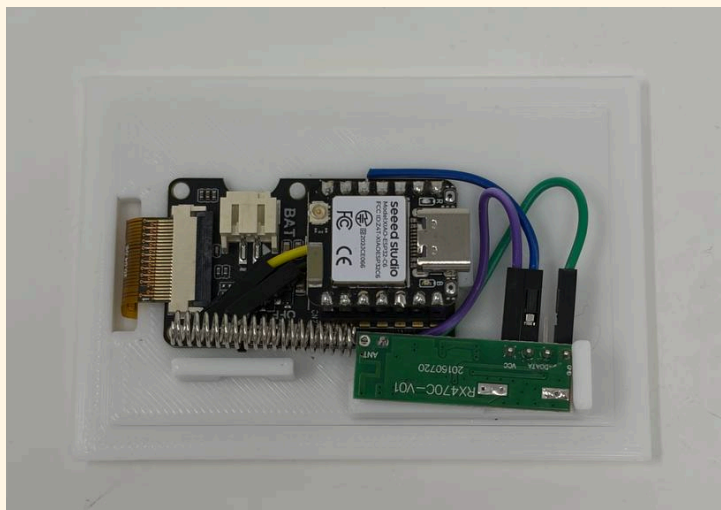


8. Connect wires to the respective inputs of the 433 MHz RF receiver.

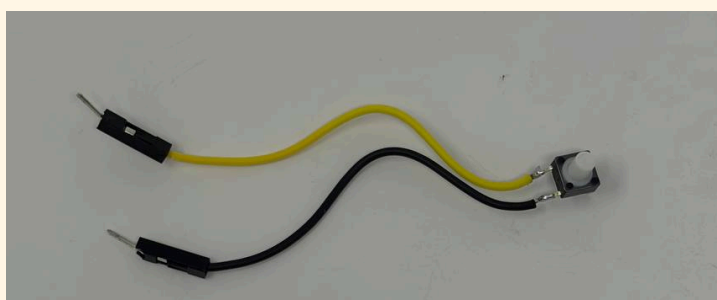
More specifically, connect D4 (blue) to the pin next to VCC, GND (green) to GND and 3V3 (purple) to VCC.



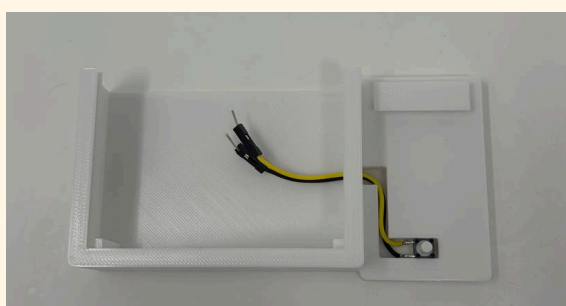
9. Connect the e-paper display to the driver board via the 24-pin ribbon cable.



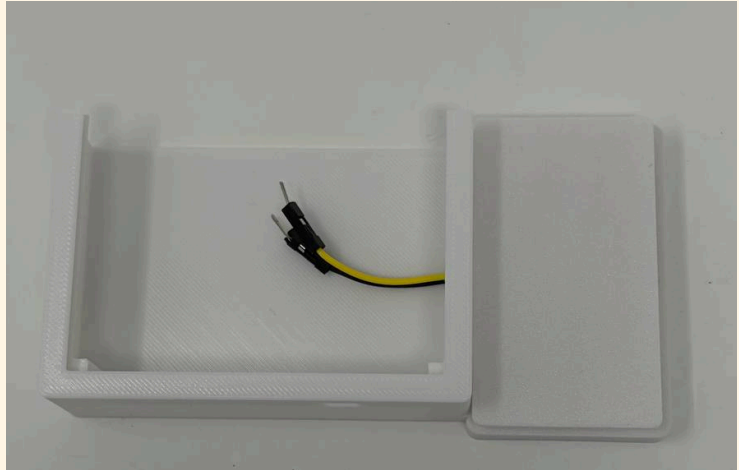
10. Solder two wires to the button.



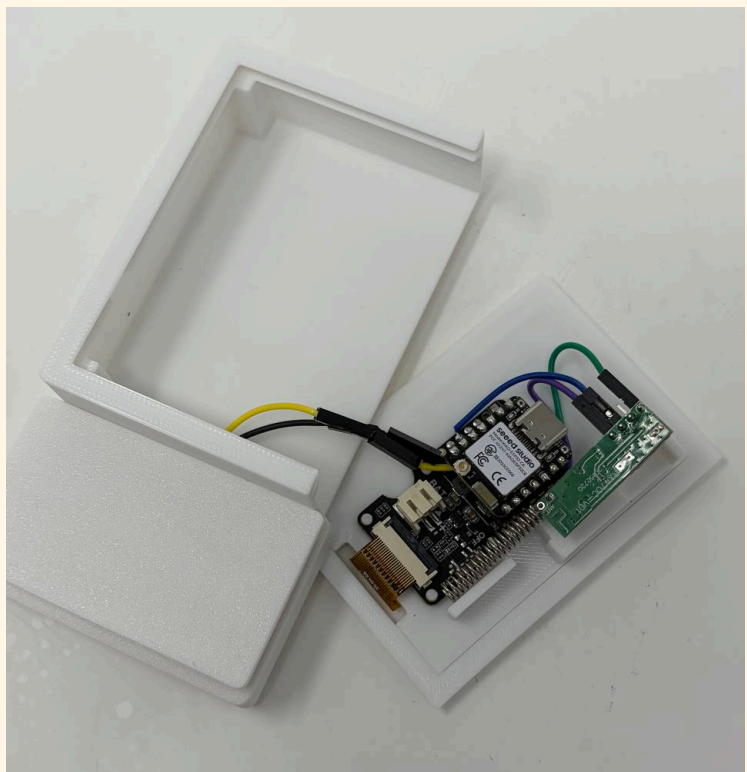
11. Insert the button into the 3D printed **Part B** slot and weave the wires through the insert so that they reach the main component.



12. Add 3D printed **Part D** followed by **Part C** to cover the wiring of the button side component of **Part B**.



13. Connect the two wires (yellow, black, from Step 10) to the ESP module's pins GND (black) and MTMS (yellow) (from Step 4).



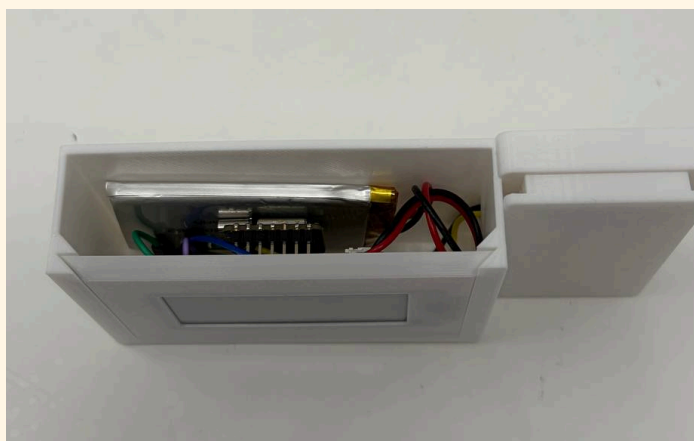
14. Slide 3D printed screen **Part G** into the slit of the box component **Part B**, ensuring that the wiring/electronic components are enclosed.



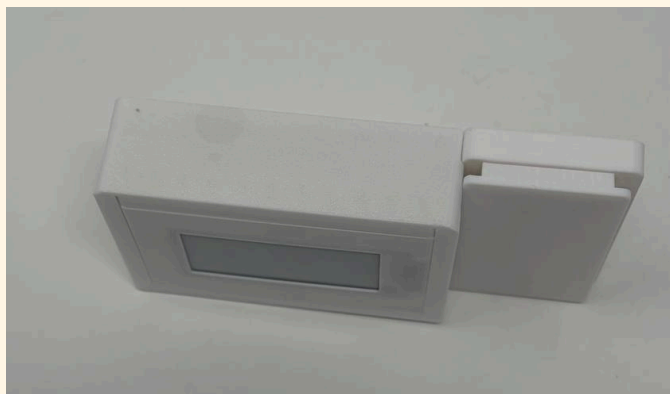
15. Connect the battery to the battery connector of the ePaper Driver board.



16. Slide the battery into the **Part B** casing.



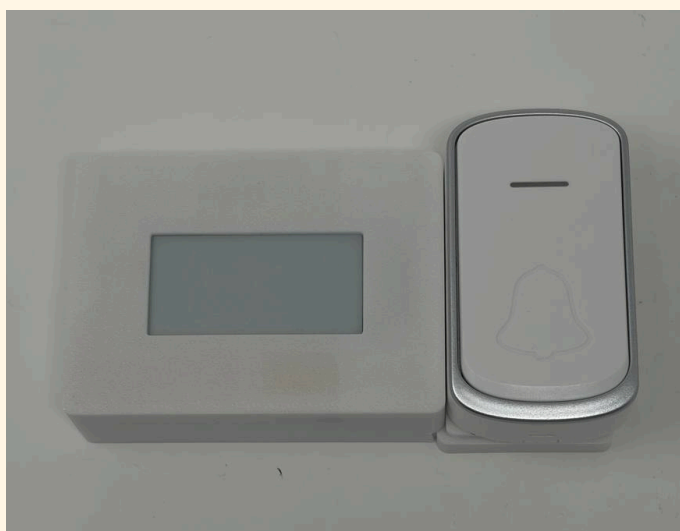
17. Slide the 3D printed lid (**Part A**) to enclose the case.



18. Slide 3D printed **Part E** on top of **Part B**.



19. Mount the Doorbell Button on top of **Part B** using double-sided tape.



20. Press the Doorbell Button. This should cause the Doorbell Indicator to light up.

If the receiver button is pressed, then the message "I'm coming" should appear on the ePaper screen.

